

INTERMEDIATE MACROECONOMICS
IS-LM MODEL OF BUSINESS CYCLES
11. IS & LM CURVES

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IS SUBMODEL

- consumer behavior: consumption = function of disposable income
- extended firm behavior: investment = function of interest rate and income
 - so far: investment was assumed to be constant
- accounting identity: income = expenditure
- IS curve: relates income Y to interest rate i when the IS submodel is in equilibrium

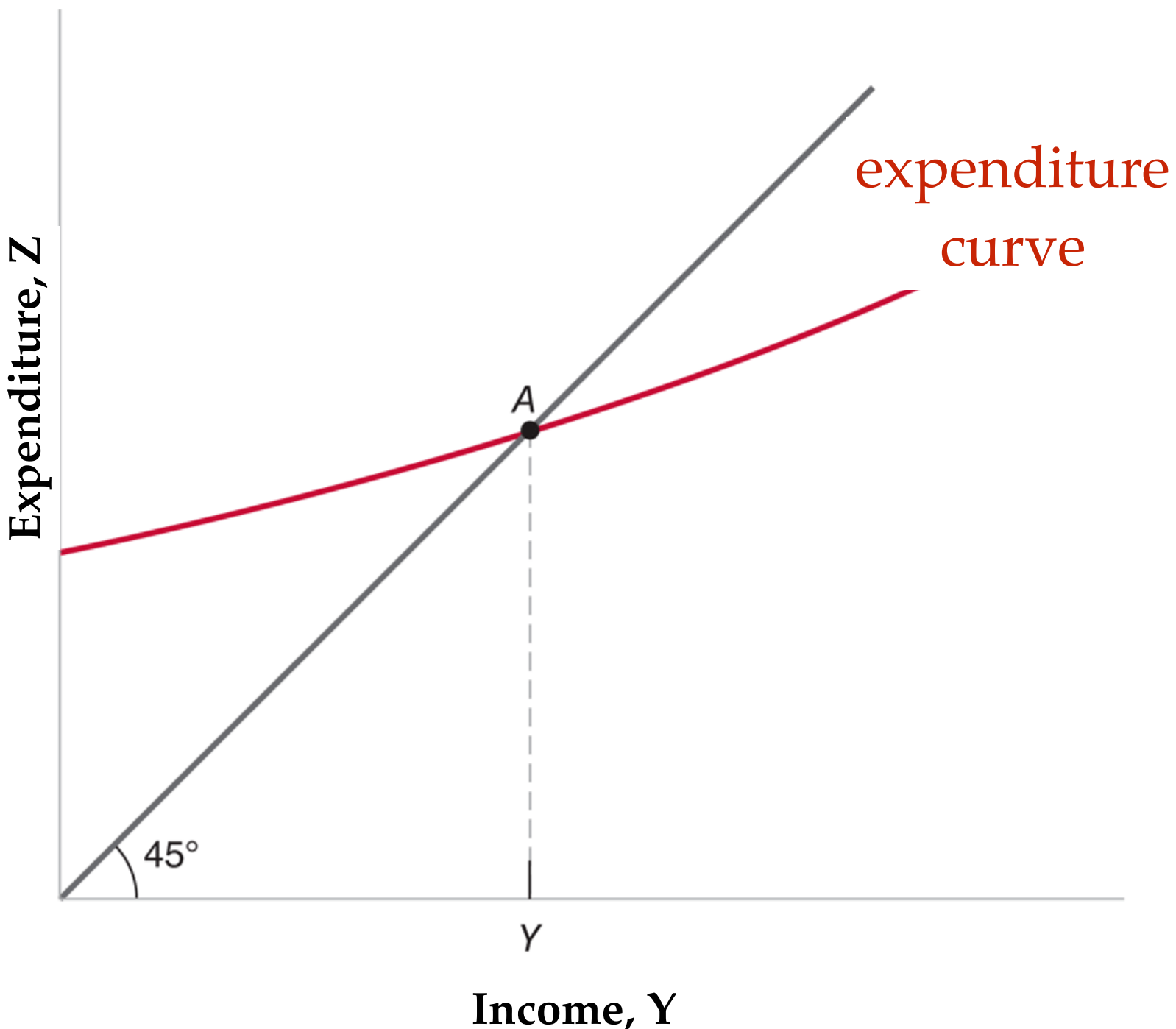
INVESTMENT

- firms invest based on income Y and interest rate i : $I = I(Y, i)$
- function $I(Y, i)$ is increasing in Y : more income means and more production by the firm
 - this justifies investment in new productive capital
- function $I(Y, i)$ is decreasing in i : higher interest rate means that it is more expensive to borrow money
 - this makes it less appealing to invest
- shape of investment function: $I(Y, i) = z_0(i) + z_1 \times Y$
 - $z_0(i)$ is a decreasing function of i
 - $z_1 < 1$ is the marginal propensity to invest out of income (MPI)

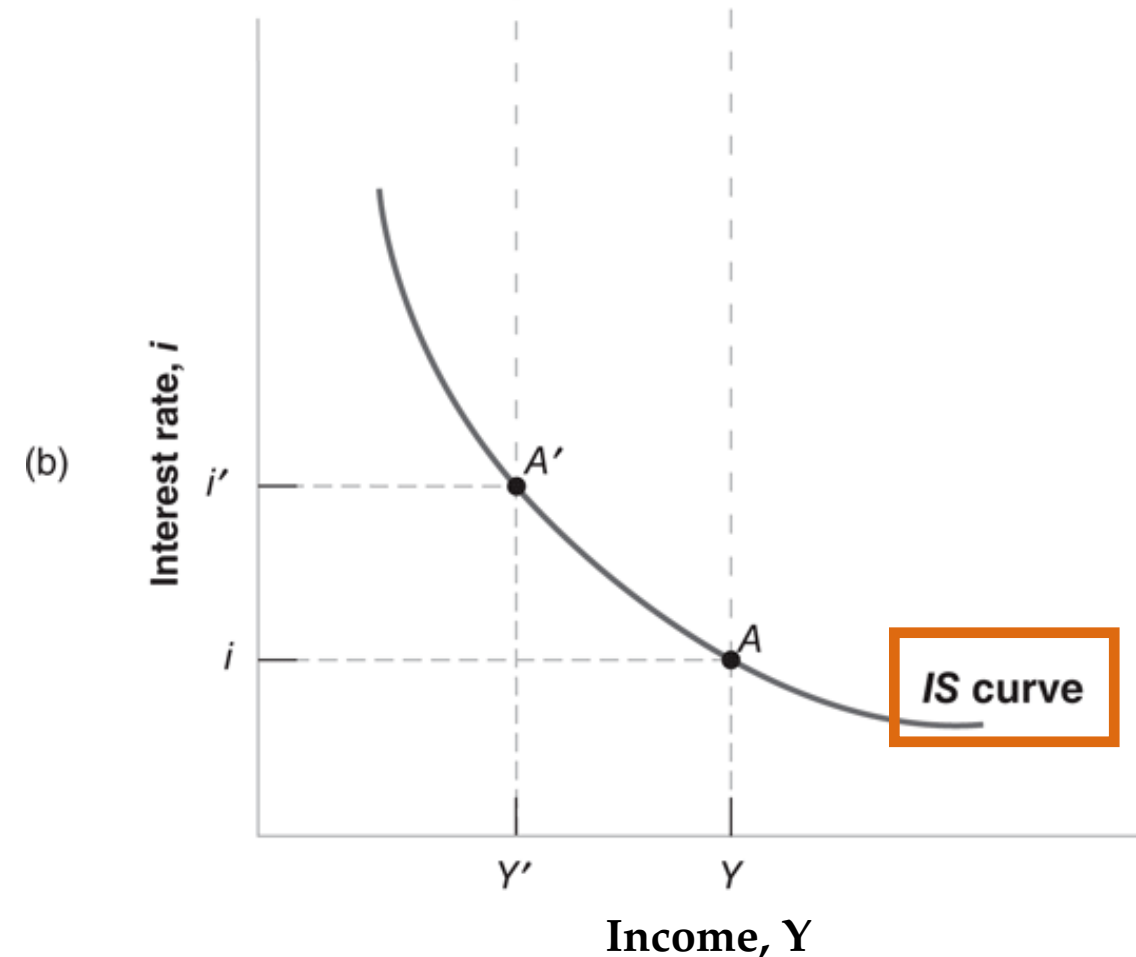
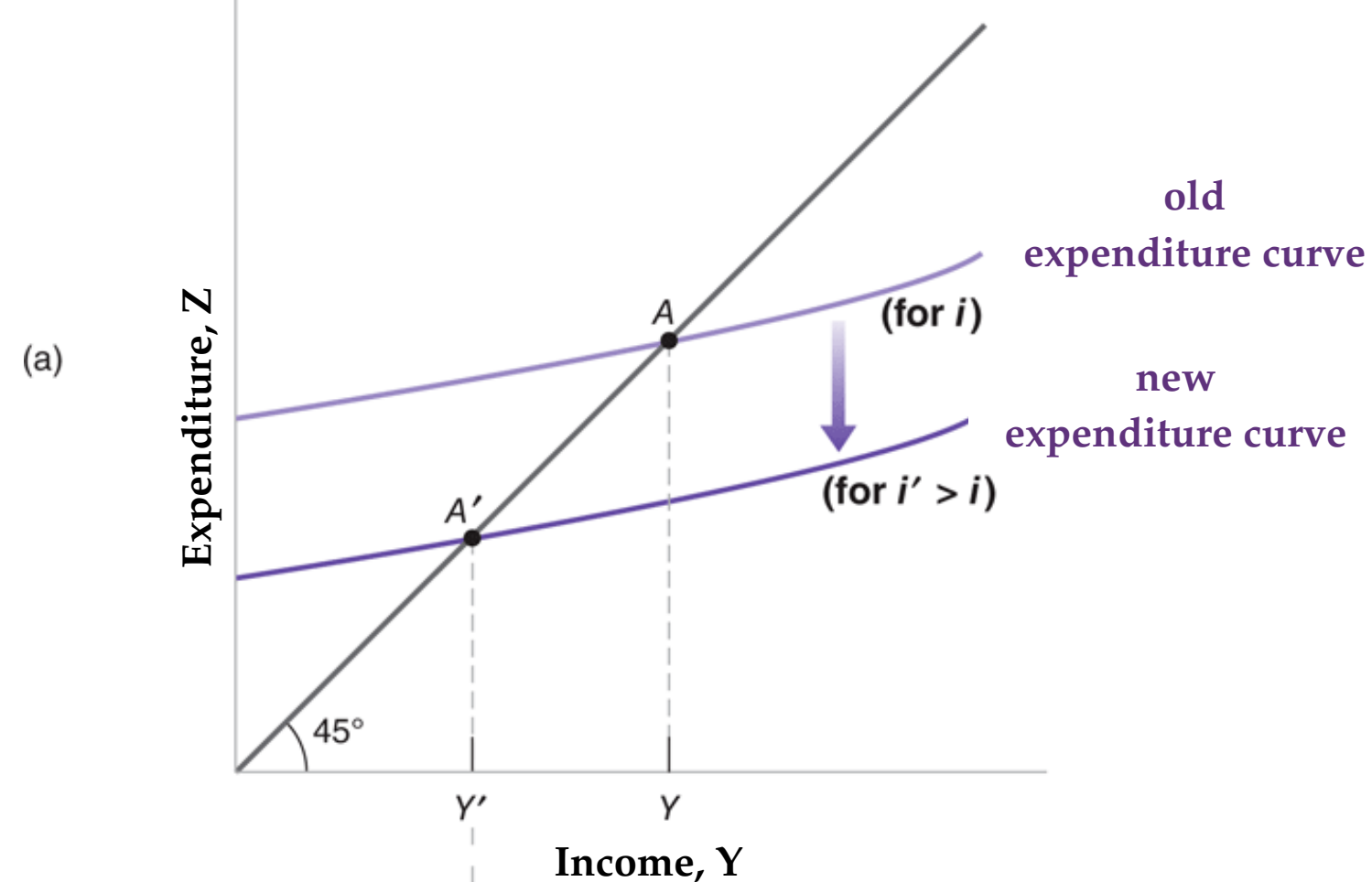
EQUILIBRIUM IN IS SUBMODEL

- expenditure function with investment:
 - $Z(Y,i) = C(Y - T) + I(Y, i) + G$
 - $Z(Y,i) = c_0 + c_1 \times (Y - T) + z_0(i) + z_1 \times Y + G$
 - $Z(Y,i) = [c_0 + z_0(i) + G - c_1 \times T] + [c_1 + z_1] \times Y$
- income = expenditure: $Z = Y$
- combining both equilibrium conditions yields $Y = Z(Y,i)$, which gives
 - $Y^* = [c_0 + z_0(i) + G - c_1 \times T] / [1 - c_1 - z_1]$
 - autonomous expenditure: $c_0 + z_0(i) + G - c_1 \times T$
 - spending multiplier: $1 / [1 - c_1 - z_1] > 1$ (need $c_1 + z_1 < 1$)
- in IS equilibrium, income Y^* depends on interest rate i

IS CURVE: GRAPHICAL CONSTRUCTION



- the expenditure curve is upward sloping
- more income Y yields more consumption C and more investment I
- the expenditure curve is flatter than 45° line
- accounting identity: expenditure $Z =$ income Y



- higher interest rate i decreases expenditure at any level of income
 - because it decreases investment $I(Y, i)$
- this leads to lower income Y
- the IS submodel implies that higher interest rate i leads to lower income Y
- IS curve is downward sloping in a (Y, i) plane

IS CURVE: NUMERICAL CONSTRUCTION

- consumers: $C(Y - T) = 30 + 0.5 \times (Y - T)$
 - marginal propensity to consume = 0.5
- firms: $I(Y, i) = 17 - 100 \times i$
- government: $G = 18$ and $T = 10$
 - government deficit: $G - T = 8$

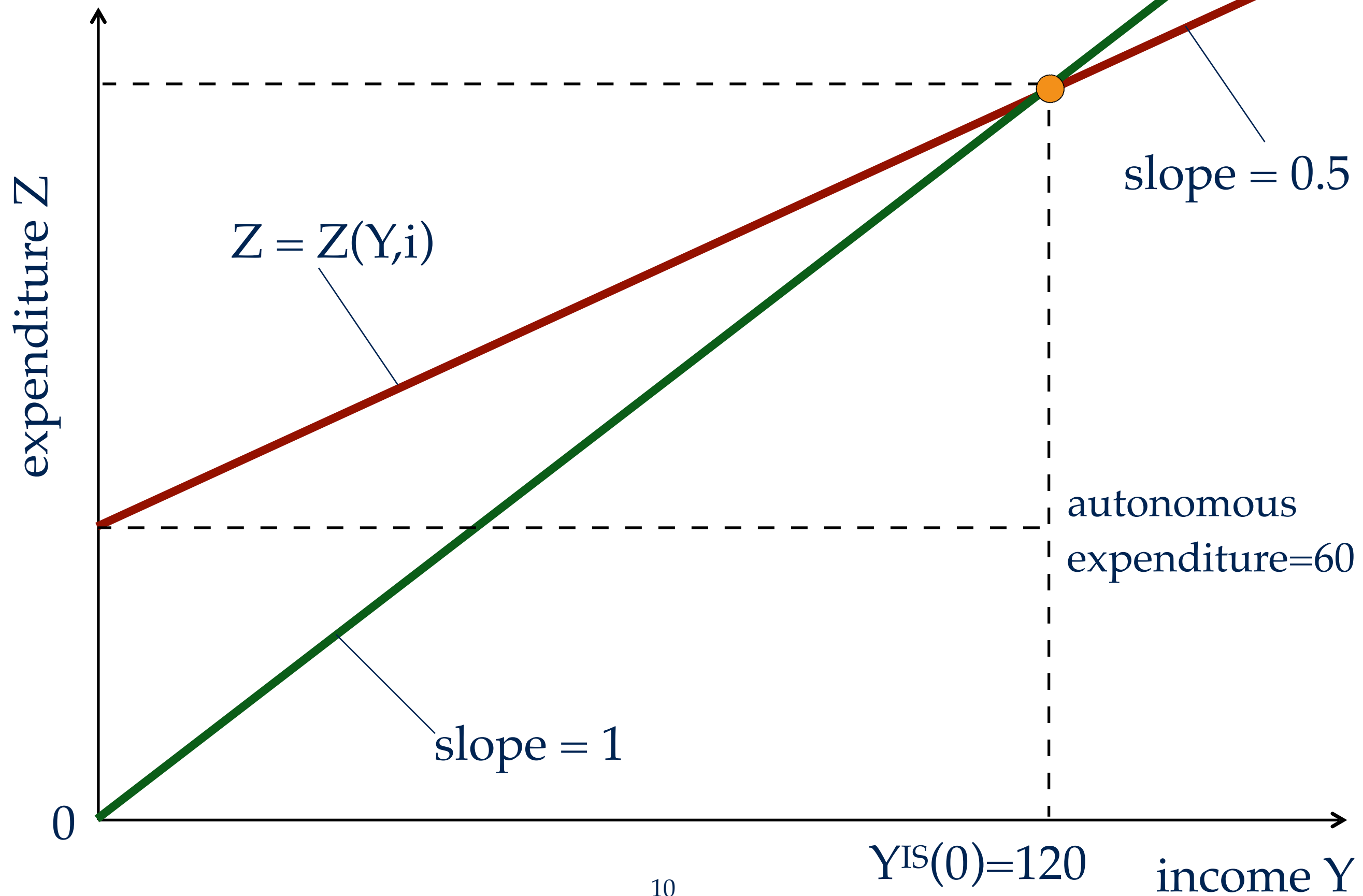
EXPENDITURE FUNCTION

- $Z(Y,i) = C(Y - T) + I(Y,i) + G$
- using our assumptions:
 - $Z(Y,i) = [30 + 0.5 \times (Y - 10)] + [17 - 100 \times i] + 18$
- reshuffling the terms to isolate Y :
 - $Z(Y,i) = [60 - 100 \times i] + 0.5 \times Y$
- autonomous expenditure: $60 - 100 \times i$
- spending multiplier: $1 / 0.5 = 2$

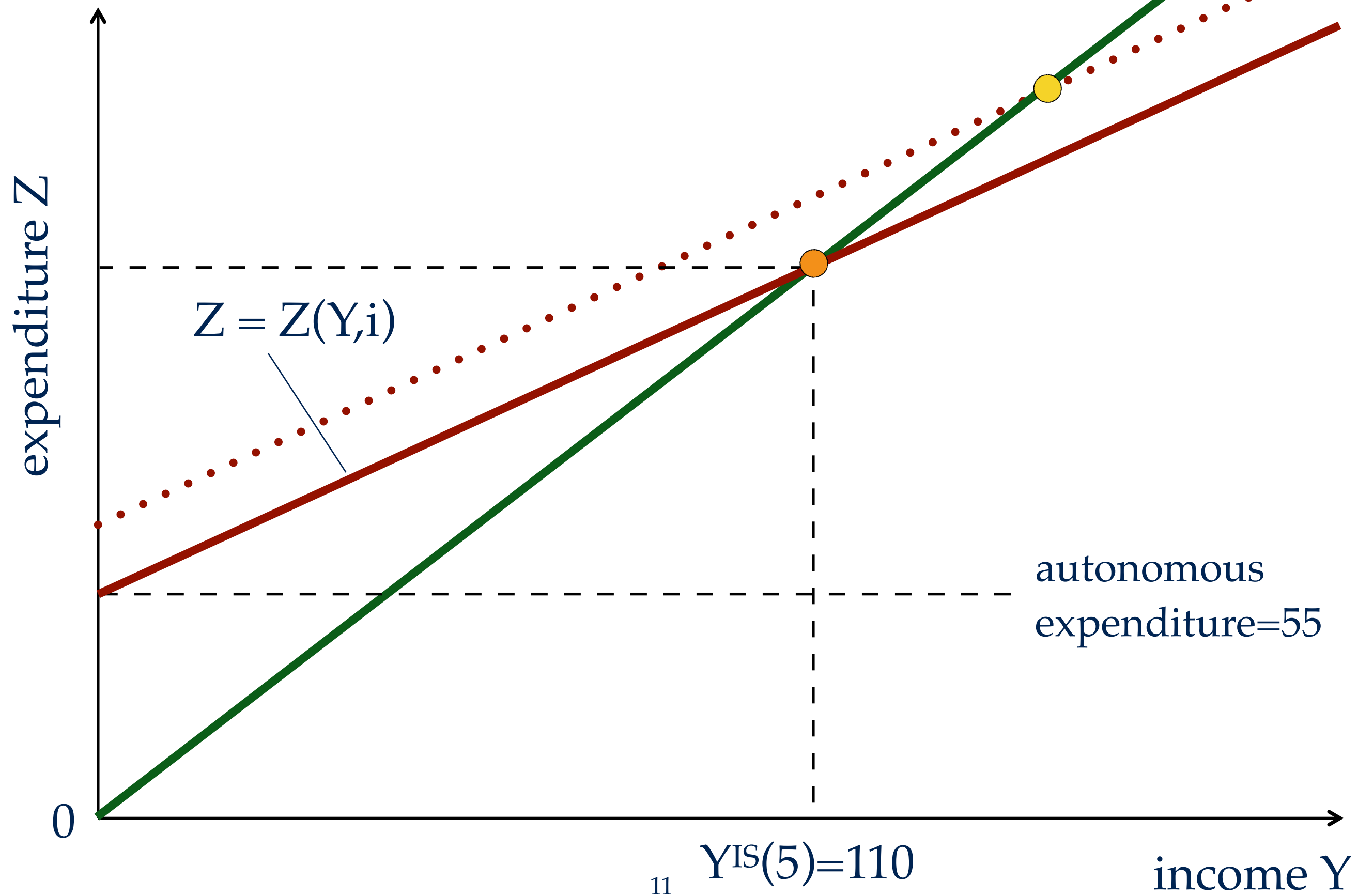
IS CURVE

- equilibrium condition 1 (accounting identity): $Y = Z$
- equilibrium condition 2 (expenditure function): $Z = Z(Y,i)$
- combining both conditions gives $Y = Z(Y,i)$, or
 - $Y = [60 - 100 \times i] + 0.5 \times Y$
 - $0.5 \times Y = [60 - 100 \times i]$
 - $Y = 2 \times [60 - 100 \times i] = 120 - 200 \times i = Y^{IS}(i)$
- the curve tracing $Y^{IS}(i)$ in a (Y,i) plane is the IS curve
 - IS curve is downward sloping because lower interest rate implies more investment and thus higher autonomous expenditure and, in equilibrium, more income

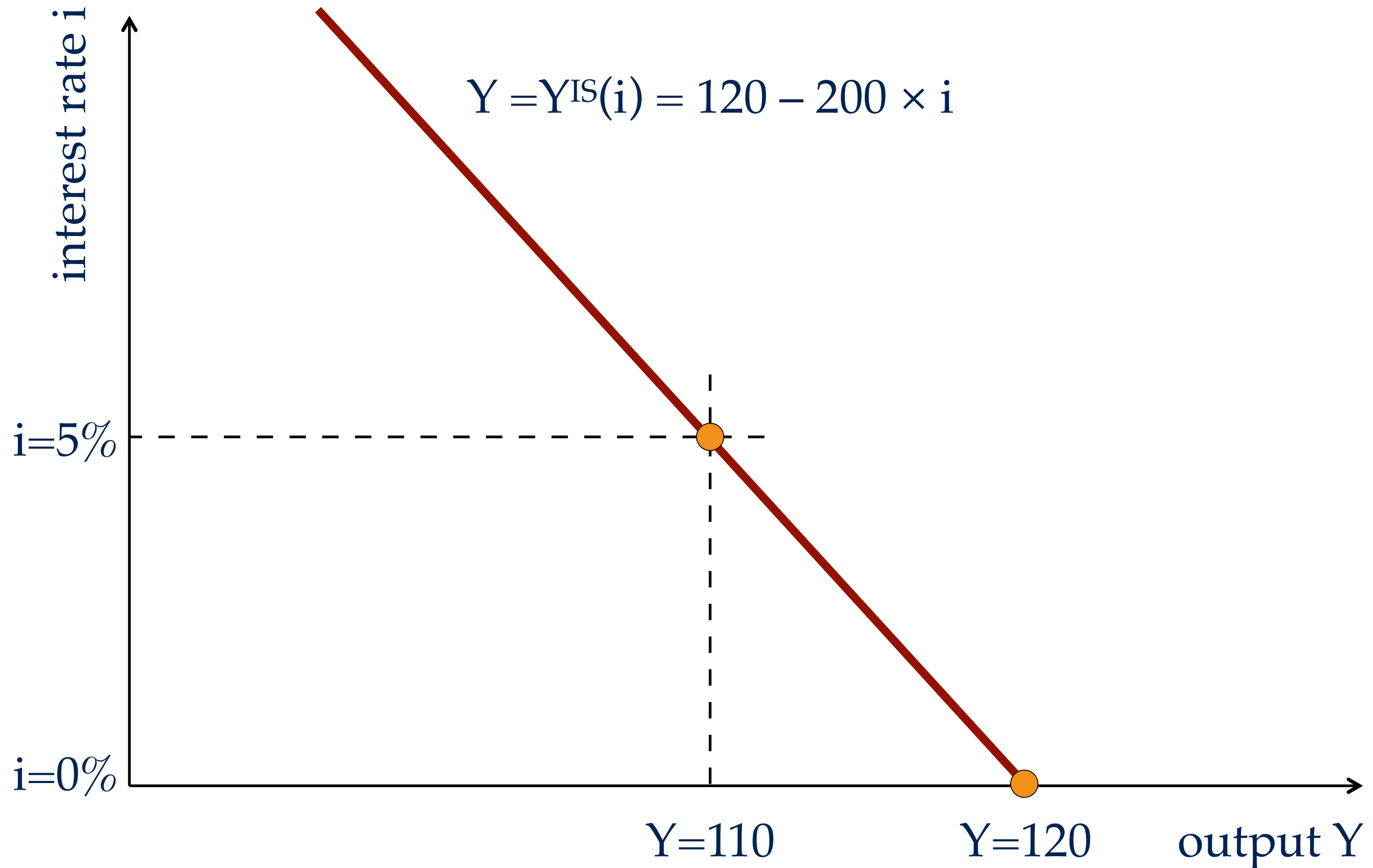
IS DIAGRAM FOR $i=0\%$



IS DIAGRAM FOR $i=5\%$



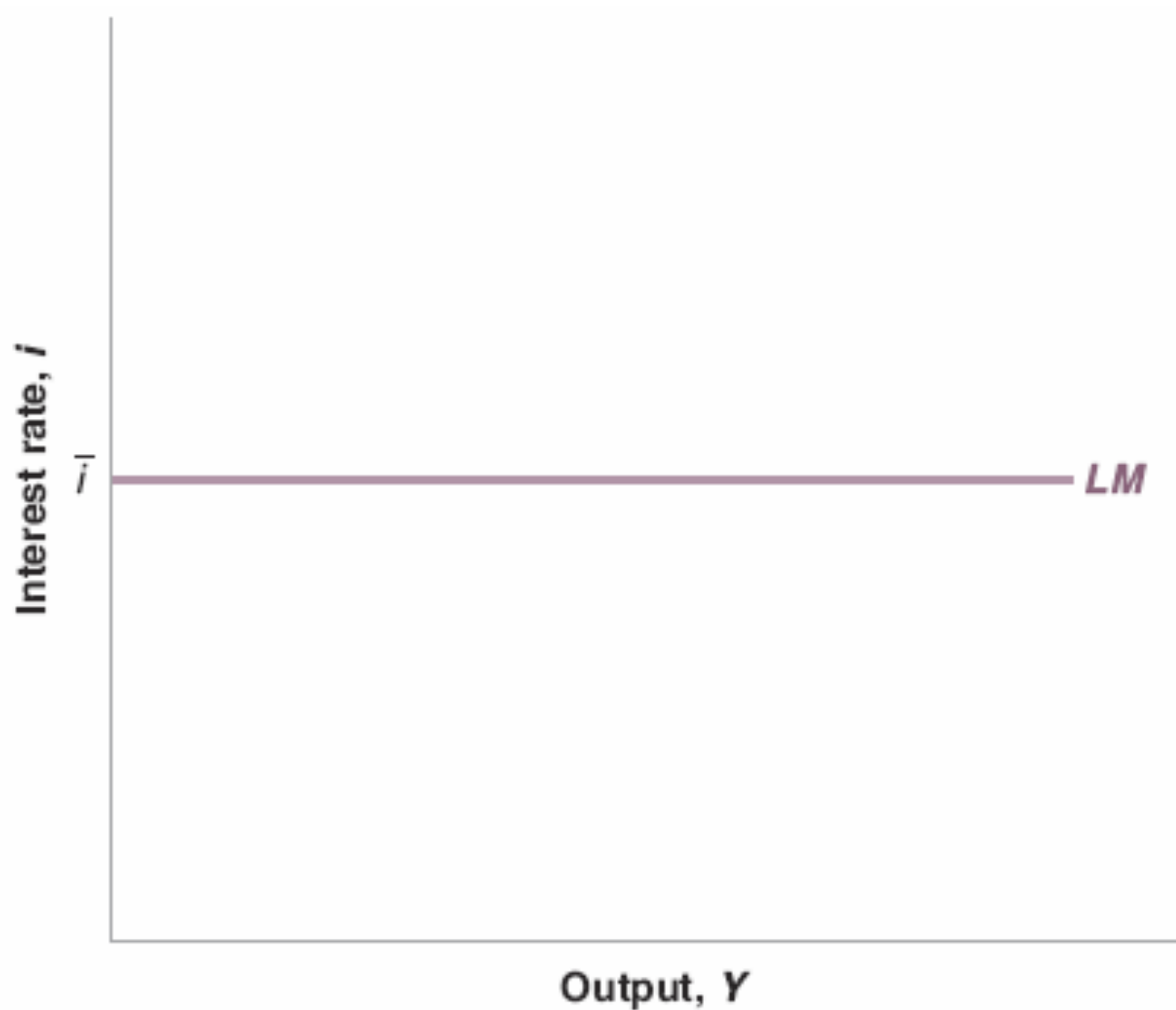
IS CURVE



LM SUBMODEL

- abstract from financial intermediaries
- demand for money: function of income + interest rate
- supply of money: determined by central bank
- in LM equilibrium: the interest rate adjusts so demand for money = supply of money
- LM curve indicates that central bank sets an interest rate i
 - in the background: central bank adjusts money supply through OMO to maintain the interest rate at i

LM CURVE



- for any level of output, the central bank maintains the interest rate at $\bar{i}$
- $\bar{i}$ must be positive: monetary policy is subject to ZLB

IS-LM EQUILIBRIUM DIAGRAM

